Semantic Segmentation of Flood Remote Sensing Images Based on Improved DeepLabv3+

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***Abstract*—To address challenges in flood remote sensing imagery such as blurred water boundaries, irregular inundation patterns, and strong background interference, this paper proposes a lightweight, enhanced DeepLabv3+ network. A progressive hollow spatial pyramid module captures multi-scale context with a stepwise expansion rate of 2 → 4 → 8 → 16. Embedded CBAM dual attention dynamically calibrates channel and spatial weights to suppress noise from tree canopies, vehicles, and other sources. Dual residual connections bridge semantic gaps between deep and shallow layers. Experiments on the FloodNet dataset demonstrate that the improved model achieves a mean IoU of 62.54%, surpassing the baseline by 2.46%. Its parameter count is reduced to 4.118 million, with GFLOPs consumption dropping to 51.991 — achieving over 80% compression compared to mainstream models. Maintaining 58 FPS, it balances accuracy and real-time performance. This research provides a lightweight and reliable semantic segmentation solution for rapid post-disaster assessment via UAVs.**

Keywords: Flood Remote Sensing; DeepLabv3+; Asynchronous Shaded Pixel Pooling; CBAM Attention

I. INTRODUCTION

In recent years, global climate change has led to frequent extreme weather events, making floods one of the primary natural disasters threatening human life and property safety [1]. Timely and accurate detection of affected areas is crucial for post-disaster emergency response, and remote sensing technology has become an essential tool for flood monitoring due to its large-scale, real-time observation capabilities [2]. However, flood remote sensing images present challenges such as blurred water boundaries, spectral similarities between inundated areas and normal water bodies, and dense small objects like building debris, making it difficult for traditional segmentation methods to achieve precise pixel-level classification [3].

Deep learning technologies offer novel solutions for remote sensing image segmentation[4]. UNet[5] demonstrated outstanding performance in medical image segmentation through its encoder-decoder architecture and skip connections, yet its shallow feature fusion mechanism exhibits limited adaptability to complex topographies. PSPNet[6] leveraged pyramid pooling modules to capture multi-scale context, but pooling operations often resulted in spatial detail loss. DeepLabv3+ [7] combines the Attenuated Spatial Pyramid Pooling (ASPP) with separable convolutions to expand the receptive field while preserving resolution, becoming a mainstream framework in semantic segmentation. However, existing methods exhibit the following shortcomings in flood scenarios: (1) The ASPP module with a fixed dilation rate struggles to adapt to the irregular shapes of flooded areas; (2) Complex background noise suppresses key features; (3) Balancing lightweight requirements with segmentation accuracy [8,9].

To address these challenges, this paper proposes an improved DeepLabv3+ algorithm for flood remote sensing image segmentation. First, a progressive ASPP structure with adjustable dilation rates (2→4→8→16) is designed to enable continuous extraction of multi-scale features. Second, a hierarchical CBAM attention mechanism is introduced to simultaneously optimize channel and spatial weight distributions during feature extraction [10]. Finally, a dual residual connection is established to promote effective fusion of deep and shallow features. Experiments on the public FloodNet dataset [11] demonstrate that the improved model achieves a 2.46% increase in mIoU compared to the original DeepLabv3+, while reducing the number of parameters, validating the effectiveness of the proposed approach.

II. METHODS AND THEORIES

This chapter proposes an improved semantic segmentation model based on the DeepLabV3+ framework. Addressing issues such as low accuracy and blurred edges in multi-scale object segmentation within flood scenarios, the model incorporates enhancements in three key areas: network architecture optimization, multi-scale feature fusion, and attention mechanisms. The overall network architecture is shown in Figure 1. It employs MobileNetV2 as the backbone network to extract multi-level features. The Progressive ASPP (Progressive Aggregated Spatial Pyramid Pooling) module enhances the fusion capability of multi-scale contextual information. Additionally, a Channel-Bidirectional Attention Mechanism (CBAM) is embedded at the end of the encoder to strengthen key feature representation. The following sections detail the design and implementation of the core improvement modules.

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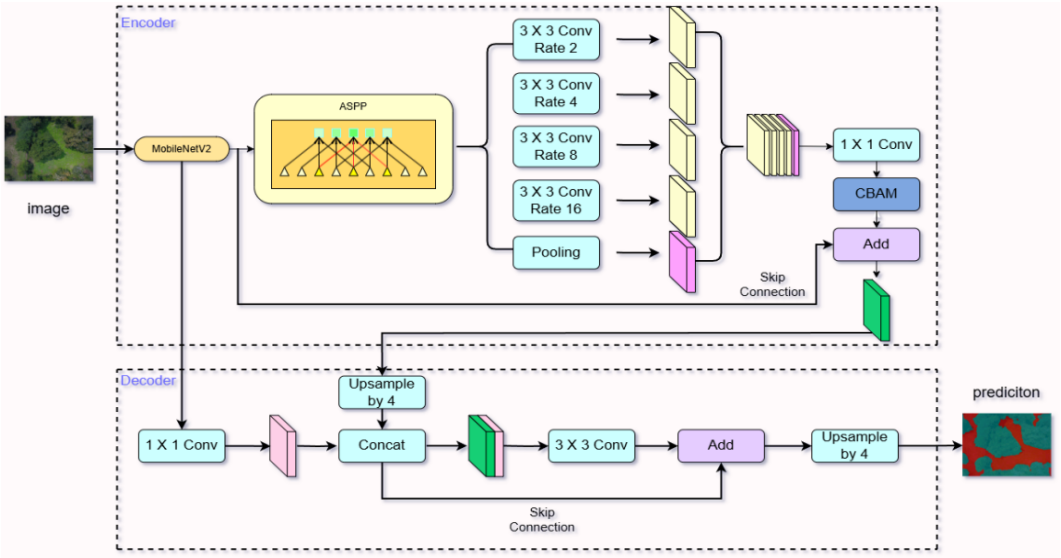


Figure 1. Schematic Diagram of the Improved DeepLabV3+ Network Architecture

The improved model adopts DeepLabV3+ as its foundational framework, comprising three components: an encoder, a multi-scale feature fusion module (Progressive ASPP), and a decoder:

(1) Encoder: Utilizes the lightweight MobileNetV2 as the backbone network. Through an inverted residual structure, it extracts shallow-level detail features (high resolution) and deep-level semantic features (low resolution), balancing computational efficiency with feature representation capability.

A common challenge in encoder-decoder architectures is the semantic gap between deep and shallow features. Deep features capture high-level semantics but lack spatial precision, while shallow features retain fine details but are semantically weak. To address this, we introduce dual residual connections between the mid-level features of the encoder (e.g., from MobileNetV2's layer3 and layer4) and the corresponding decoder layers. These connections are implemented as element-wise additions after 1×1 convolutions to align channel dimensions. This design promotes the fusion of spatial details and semantic information, reduces information loss during upsampling, and facilitates gradient propagation during training. As a result, the model achieves clearer object boundaries and improved consistency in segmenting irregular flood regions.

(2) Multi-scale Feature Fusion: A Progressive ASPP module is designed at the encoder's end. It extracts multi-scale contextual information through dilated convolutions with stepwise dilation rates, while residual connections prevent information loss.

The original DeepLabv3+ employs an Atrous Spatial Pyramid Pooling (ASPP) module with fixed dilation rates (e.g., 6, 12, 18) to capture multi-scale context. However, in flood remote sensing imagery, water bodies and inundated areas exhibit highly irregular shapes and scale variations. A fixed set of dilation rates may not align well with the natural scale distribution of flood features.

To better adapt to the continuous scale variation of flood objects, we propose a Progressive ASPP module with dilation rates increasing in a power-of-two sequence: 2, 4, 8, 16. This design is motivated by the typical scale distribution of flood features observed in the FloodNet dataset, where water boundaries and connected components often span multiple scales in a near-exponential manner. The progressive structure allows the network to capture context from finer local details (rate=2) to broader regional contexts (rate=16) in a smooth and continuous manner, reducing the risk of grid artifacts caused by large and abrupt dilation changes. Each branch in the Progressive ASPP incorporates a residual connection to preserve gradient flow and feature integrity. The outputs of all branches are concatenated and fused via a 1×1 convolution.

(3) Attention Enhancement: A CBAM module is inserted after ASPP output. It dynamically calibrates feature map weights via channel and spatial attention, suppressing background noise while amplifying responses in target regions.

III. Experiments and Results Analysis

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A. Experimental Environment and Hyperparameter Settings

The experiment was built using PyTorch 1.13.0, with Python 3.8 as the scripting language, CUDA 11.7 drivers, and Windows 11 as the operating system. Hardware platform: AMD Ryzen 9 7940HS (4.0 GHz), 32 GB DDR5 memory, RTX 4060 laptop GPU (8 GB VRAM).

The network adopts the DeepLabV3+ architecture, utilizing a pre-trained MobileNetV2 backbone from ImageNet with an input resolution of 512×512. Training occurs in two phases: the first 50 epochs freeze the backbone while updating only the decoder, with batch size=8; the subsequent 50 epochs perform end-to-end fine-tuning with batch size=4. The optimizer is Adam with an initial learning rate of 5×10^{-4}, cosine annealed to 1×10^{-6}, and weight decay of 1×10^{-4}. The loss function is a weighted blend of cross-entropy and Dice loss. Total training spans 100 epochs with early stopping at patience=10.

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B. Dataset

The experiment employed the FloodNet high-resolution post-flood UAV imagery dataset for model training and evaluation. FloodNet was released by the Bina Lab team at the University of Maryland, Baltimore County (UMBC). The original dataset comprises 2,343 images with a spatial resolution of approximately 2 cm, covering 10 post-disaster scene elements including buildings, roads, water bodies, and vehicles. It features pixel-level dense annotations, with representative examples shown in Figure 2. The dataset was divided into training, validation, and test sets in a 6:2:2 ratio, containing 1,406, 469, and 468 images respectively. This ensures that the same area or adjacent swaths do not appear across sets, minimizing the risk of leakage due to spatial autocorrelation.



Figure 2. FloodNet Dataset

C. EquationsEvaluation Indicators

The experiment comprehensively evaluates the model using mean intersection over union (mIoU), Params(M), GFLOPS, and FPS. Here, mIoU denotes the average intersection over union ratio between predicted regions and ground truth regions across all categories, calculated as follows:

$$mIou = \frac{1}{k+1}\sum_{i=0}^{k}\frac{pii}{\sum_{j=0}^{k} pij + \sum_{j=0}^{k} pji - pii} \quad (1)$$

D. Analysis of Experimental Results

1) Melting Experiment

To validate the contribution of each module to DeepLabV3+ segmentation performance, four sets of ablation experiments were designed, with results shown in Table 1. Table 1 verifies the contribution of each module: The baseline DeepLabV3+ (MobileNet) achieved mIoU 60.37% with 5.815 M parameters. Replacing it with progressive ASPP increased parameters by only 0.009 M while improving mIoU by 1.33 percentage points; Embedding CBAM further reduced parameters to 4.11M while achieving mIoU 61.58%.Adding dual residual connections alone improved mIoU to 61.92% with negligible parameter increase. Combining both modules yielded mIoU 62.54%, representing a cumulative 2.17 percentage point improvement. Parameters and computational cost decreased by 29.2% and 1.8% respectively, while maintaining 58 FPS—sufficient for real-time flood analysis. This demonstrates the complementary nature of both modules without introducing additional latency.

TABLE I. ABLATION EXPERIMENTS

Model	Miou(%)	Params(M)	GFLOPS	FPS
DeepLab	60.37	5.815	52.934	58
+ASPP	61.70	5.824	52.935	58
+CBAM	61.58	**4.11**	**51.991**	**57**
+Dual Residual	61.92	5.820	52.934	58
Ours	**62.54**	4.118	**51.991**	58

2) Comparative Trial

Table 2 presents a comprehensive comparison between mainstream models on FloodNet and the proposed method. U-Net, PSPNet, and DeepLabV3+Xception achieve mIoU values of 55.2%, 57.8%, and 58.23%, respectively, but their parameter counts reach 31.2M, 46.7M, and 25.4M, with GFLOPs of 120.4, 150.2, and 89.6, making deployment on drone edge devices challenging. Our model, based on DeepLabV3+ lightweight backbone and integrating Progressive ASPP with CBAM, achieves 62.54% mIoU with only 4.118M parameters and 51.991 GFLOPs—a 4.31 percentage point improvement over the best baseline model. with parameter and computational complexity reduced by 84% and 42%, respectively. This demonstrates its comprehensive advantages in accuracy, efficiency, and real-time performance, meeting the rapid analysis demands of mobile platforms in post-flood disaster scenarios.

TABLE II. COMPARATIVE EXPERIMENTS

Model	Miou(%)	Params(M)	GFLOPS
U-net	55.2	31.2	120.4
PSPNet	57.8	46.7	150.2
Deeplab+Xception	58.23	25.4	89.6
Ours	**62.54**	**4.118**	**51.991**

3) Visualization results

To validate the model's detection performance on the dataset, testing was conducted using the FloodNet dataset. Selected visualization results are shown in Figure 3. The figure presents both the original imagery and the prediction results from the proposed model, clearly illustrating the distinction between buildings and small water bodies. The proposed model produces smoother mask edges with significantly fewer missed and false-positive pixels, further demonstrating its robustness in complex post-disaster scenarios.

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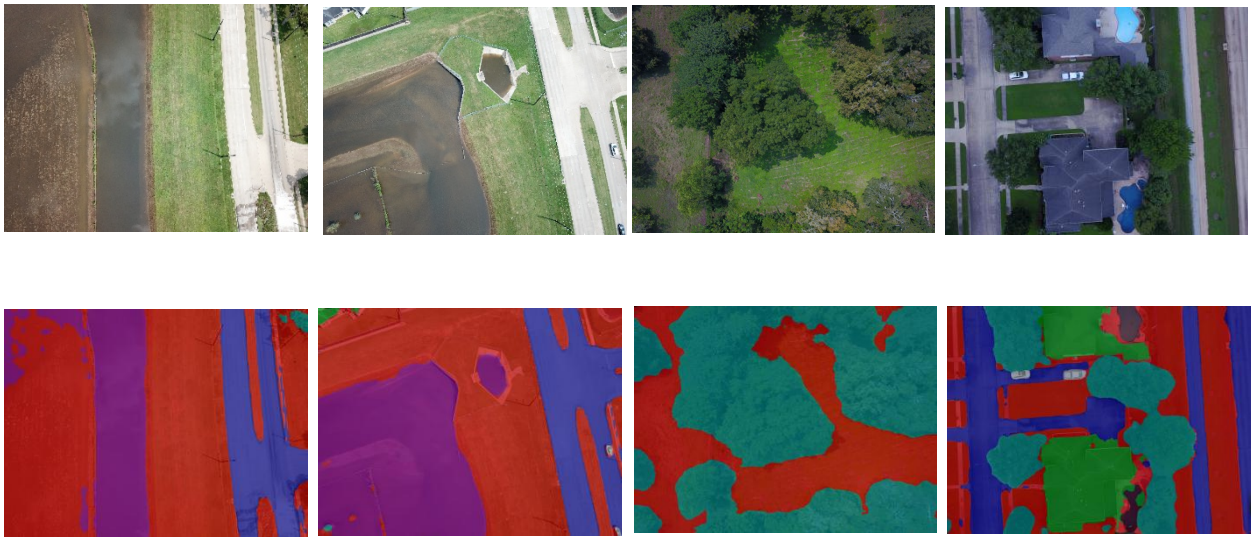


Figure 3. Visualization Results

IV. Conclusion

This paper addresses challenges in multi-scale target recognition, edge blurring, and backgr8ound interference during remote sensing image segmentation for flood detection. We propose an improved semantic segmentation model based on the DeepLabv3+ framework. By incorporating Progressive ASPP (Progressive Aggregated Sparse Pyramid Pooling), the CBAM attention mechanism, and dual residual connections, the model achieves enhanced segmentation accuracy and robustness in complex flood scenarios. Experiments on the FloodNet public dataset demonstrate that our method achieves a 2.46% improvement in mIoU compared to the original DeepLabv3+, while simultaneously reducing the number of model parameters. This validates the dual advantages of the proposed improvements in both accuracy and lightweight performance.

Although our method achieves promising results in flood image segmentation, limitations remain: limited adaptability to extreme lighting conditions and room for improvement in segmenting extremely small objects (e.g., isolated building debris). Future work will explore more efficient feature fusion strategies and incorporate structures like Transformers to enhance long-range dependency modeling capabilities, thereby further improving the model's generalization performance in complex disaster scenarios.

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